Software Requirements Specification for

Digital Wallet Management System

Version 1.0

Approved

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Revision History

Name	Date	Reason For Changes	Version
B.Tech Student	02-Sep-2026	Initial draft of SRS	1.0

1.Introduction

a. Purpose

This Software Requirements Specification (SRS) document describes the functional and non-functional requirements for the Digital Wallet Management System (DWMS), Version 1.0. The document covers the complete software product that enables users to create and manage digital wallets, perform secure fund transfers, make payments, view transaction history, and manage account settings. This SRS serves as the primary reference for developers, testers, project managers, and stakeholders throughout the software development lifecycle.

b. Document Conventions

The following conventions are used throughout this document:

- Requirement identifiers follow the format REQ-X.Y (e.g., REQ-4.1.1), where X denotes the system feature and Y the specific requirement number.
- Priority levels are indicated as High, Medium, or Low.
- Bold text is used for emphasis and for requirement identifiers.
- is used for notes, examples, and explanatory remarks.
- Higher-level requirements are assumed to be inherited by detailed requirements unless explicitly stated otherwise.

c.Intended Audience and Reading Suggestions

This document is intended for the following audiences:

- Developers: Should read the entire document, focusing on Sections 3, 4, and 5 for interface and functional requirements.
- Project Managers: Should review Sections 1, 2, and 5 for scope, constraints, and non-functional requirements.
- Testers: Should focus on Sections 4 and 5 to derive test cases from functional and non-functional requirements.
- Users / Stakeholders: May read Sections 1, 2.1– 2.3, and 4 for an overview of product capabilities.
- Documentation Writers: Should refer to Section 2.6 and the feature descriptions in Section 4.

d. Product Scope

The Digital Wallet Management System is a web and mobile-accessible application that allows

registered users to:

- Create and manage personal digital wallets linked to bank accounts or cards. • Add money (top-up) to the wallet from linked bank accounts or cards.
- Transfer funds to other wallet users or bank accounts.
- Make payments to merchants and service providers.
- View real-time balance and detailed transaction history.
- Receive notifications for transactions and account activities.

The system aims to provide a secure, convenient, and user-friendly alternative to traditional cash and card payments, supporting financial inclusion and digital payment adoption. It is designed as a standalone product that can integrate with existing banking APIs and payment gateways.

e. References

1. IEEE Std 830-1998 – IEEE Recommended Practice for Software Requirements Specifications.
2. Reserve Bank of India (RBI) Guidelines on Prepaid Payment Instruments (PPIs). 3. Payment Card Industry Data Security Standard (PCI-DSS) v4.0.
4. OWASP Top 10 Web Application Security Risks.
5. UML 2.5 Specification – Object Management Group.

2. Overall Description

a. Product Perspective

The Digital Wallet Management System is a new, self-contained product developed as an academic project. It is not a replacement for an existing system but is designed to interoperate with external banking systems and payment gateways. The system consists of the following major components:

- Client Applications: Web application (responsive) and optional Android/iOS mobile apps.
- Backend Server: RESTful API services handling business logic, authentication, and transaction processing.
- Database: Relational database storing user profiles, wallet balances, and transaction records.
- External Interfaces: Integration with banking APIs (for fund transfer), payment gateways, SMS/email gateways for notifications, and KYC verification services.

b. Product Functions

The major functions of the Digital Wallet Management System are summarized below:

- User Management: Registration, login, profile management, KYC verification, and password recovery.
- Wallet Operations: Create wallet, view balance, add money (top-up), withdraw to bank account.
- Payments & Transfers: Peer-to-peer (P2P) transfers, merchant payments, bill payments, and QR-code based payments.
- Transaction Management: View transaction history, filter/search transactions, generate statements, and dispute handling.
- Notifications: Real-time alerts via push notification, SMS, and email for transactions and security events.

- Administration: User management, transaction monitoring, fraud detection support, and system configuration by administrators.

c. User Classes and Characteristics

The system supports the following user classes:

- Registered User (Primary): Individuals who create an account, complete

KYC, and use the wallet for daily transactions. They are expected to have

basic smartphone/computer literacy. Frequency of use: daily to weekly.

- Merchant User: Business entities that accept payments via the wallet. They require additional merchant onboarding and settlement features. Technical expertise: moderate.
- Administrator: System operators who manage users, monitor transactions, handle disputes, and configure system parameters. High technical expertise and elevated privileges.
- Guest / Unregistered Visitor: Can view public information and registration pages only. Limited access.

d. Operating Environment

The software will operate in the following environment:

- Client Side: Modern web browsers (Chrome, Firefox, Edge, Safari – latest two versions) on Windows, macOS, Linux, Android, and iOS. Responsive design for mobile devices.
- Server Side: Linux-based server (Ubuntu 22.04 LTS or equivalent) running Node.js / Java / Python backend, with Nginx or Apache as reverse proxy.
- Database: PostgreSQL 15+ or MySQL 8.0+.
- Network: Internet connectivity with HTTPS (TLS 1.2 or higher) required for all communications.

e. Design and Implementation Constraints

- Must comply with RBI guidelines for Prepaid Payment Instruments (PPIs) regarding KYC, transaction limits, and fund loading.
- All sensitive data (passwords, card details, PINs) must be encrypted at rest and in transit.
- The system must use HTTPS exclusively; HTTP shall be redirected. • Database and application servers should support horizontal scaling for future growth. • Third-party payment gateway and banking API availability and rate limits are external constraints.
- Development shall follow secure coding practices aligned with OWASP guidelines.

f. User Documentation

The following documentation will be delivered with the software:

- Online Help / FAQ section within the application.
- User Manual (PDF) covering registration, wallet operations, and troubleshooting. • Admin Guide for system administrators.
- In-app contextual tooltips and onboarding walkthrough for first-time users.

g. Assumptions and Dependencies

Assumptions:

- Users possess a valid mobile number and email address for registration and OTP verification.
- Users have access to a smartphone or computer with internet connectivity. • Banking and payment gateway partners will provide stable APIs for integration. • KYC documents

(Aadhaar, PAN, etc.) can be verified through third-party services or manual review.

Dependencies:

- Availability of SMS and email gateway services for OTP and notifications. • Reliable internet connectivity between the application server and external banking/payment systems.
- Compliance with applicable data protection and financial regulations in the target jurisdiction.

3. External Interface Requirements

a. User Interfaces

The system shall provide a clean, intuitive, and responsive graphical user interface with the following characteristics:

- Consistent navigation bar / bottom navigation with access to Home, Wallet, Payments, History, and Profile.
- Dashboard displaying current wallet balance, recent transactions, and quick-action buttons (Add Money, Send, Pay).
- Form validation with clear error messages displayed near the relevant fields. • Support for both light and dark themes (optional).
- Accessible design following WCAG 2.1 Level AA guidelines where feasible. • Confirmation dialogs for critical actions (send money, withdraw, change password).

b. Hardware Interfaces

The system has minimal direct hardware dependencies:

- Client devices: Smartphones, tablets, or computers with a modern browser and internet connectivity.
- Optional: Camera access for QR-code scanning (payment and KYC document capture).
- Server hardware: Standard cloud or on-premise servers capable of running the chosen backend stack and database.

c. Software Interfaces

The system shall interface with the following software components:

- Database: PostgreSQL/MySQL for persistent storage of users, wallets, and transactions.
- Payment Gateway API: For card/net-banking top-ups and merchant settlements (e.g., Razorpay, Stripe, or equivalent).
- Banking APIs / UPI: For account verification, fund transfers, and withdrawals (where available).
- SMS Gateway: For OTP delivery and transaction alerts.
- Email Service: For account verification, statements, and notifications. • KYC Verification Service: Third-party or government APIs for identity verification.

d. Communications Interfaces

- All client-server communication shall use HTTPS (TLS 1.2+).
- RESTful JSON APIs for data exchange between frontend and backend.
- WebSocket or Server-Sent Events may be used for real-time balance and notification updates.

- OTP and alerts delivered via SMS and email.
- API authentication via JWT (JSON Web Tokens) or equivalent secure token mechanism.

4. System Features

This section organizes the functional requirements by major system features.

a. User Registration and Authentication

i. Description and Priority

This feature enables new users to register, verify their identity, and securely log into the system. Existing users can recover forgotten passwords. Priority: High.

ii. Stimulus/Response Sequences

1. Registration: User enters mobile number/email → System sends OTP → User verifies OTP → User sets password and basic details → System creates account and prompts for KYC.
2. Login: User enters credentials → System validates → Optional 2FA/OTP → User is granted access to dashboard.
3. Password Recovery: User requests reset → System sends OTP/link → User sets new password.

iii. Functional Requirements

REQ-4.1.1: The system shall allow users to register using a unique mobile number and/or email address. REQ-4.1.2: The system shall verify the mobile number/email via a time-bound OTP (valid for 5– 10 minutes). REQ-4.1.3: The system shall enforce strong password rules (minimum 8 characters, combination of letters, numbers, and special characters).

REQ-4.1.4: The system shall support multi-factor authentication (OTP or authenticator app) for login. REQ-4.1.5: The system shall lock the account temporarily after a configurable number of consecutive failed login attempts.

REQ-4.1.6: The system shall provide a secure password reset mechanism using OTP or email link.

b. Wallet Management

i. Description and Priority

This feature allows users to create a digital wallet, view balance, add funds (top-up), and withdraw money to a linked bank account. Priority: High.

ii. Stimulus/Response Sequences

1. Add Money: User selects amount and payment method → System redirects to payment gateway → On success, wallet balance is updated and confirmation is shown.
2. Withdraw: User enters amount and bank account → System validates limits and balance → Initiates transfer → Updates balance and notifies user.

iii. Functional Requirements

REQ-4.2.1: The system shall create a unique wallet for each successfully registered and KYC-verified user. REQ-4.2.2: The system shall display the current wallet balance in real time on the dashboard.

REQ-4.2.3: The system shall allow users to add money to

the wallet via linked bank account, debit/credit card, or

UPI. REQ-4.2.4: The system shall enforce configurable

daily/monthly top-up and withdrawal limits as per

regulatory guidelines.

REQ-4.2.5: The system shall allow users to link one or more bank accounts after verification.
REQ-4.2.6: The system shall prevent wallet balance from going negative under any circumstance.

c. Fund Transfer and Payments

i. Description and Priority

This feature enables peer-to-peer transfers, merchant payments, and QR-code based payments. Priority: High.

ii. Stimulus/Response Sequences

1.P2P Transfer: User enters recipient mobile/UPI ID and amount → System shows recipient name for confirmation → User authenticates (PIN/OTP) → Funds are transferred → Both parties receive notification.

2. QR Payment: User scans merchant QR → Amount is auto-filled or entered → User confirms and authenticates → Payment is completed.

iii. Functional Requirements

REQ-4.3.1: The system shall allow users to transfer funds to other registered wallet users using mobile number or unique wallet ID.

REQ-4.3.2: The system shall require secondary authentication (PIN,

biometric, or OTP) for all fund transfers and payments above a configurable

threshold. REQ-4.3.3: The system shall support QR-code based payments

for both generation (receive) and scanning (pay).

REQ-4.3.4: The system shall display a confirmation screen

with recipient details and amount before finalizing any

transfer. REQ-4.3.5: The system shall process successful

transfers atomically (debit and credit must both succeed or

both fail).

REQ-4.3.6: The system shall generate a unique transaction reference ID for every transfer and payment.

d. Transaction History and Reporting

i. Description and Priority

Users can view, filter, and export their transaction history. Priority: Medium.

ii. Stimulus/Response Sequences

1.User opens History → System displays paginated list of transactions sorted by date (newest

first) → User applies filters (date range, type, status) → System returns filtered results → User may download statement as PDF/CSV.

iii. Functional Requirements

REQ-4.4.1: The system shall maintain a complete, immutable record of all transactions for each user. REQ-4.4.2: The system shall allow users to filter transactions by date range, transaction type (credit/debit), and status.

REQ-4.4.3: The system shall provide a downloadable transaction statement in PDF and CSV formats for a selected period. REQ-4.4.4: The system shall display transaction status (Success, Pending, Failed) clearly for each entry.

e. Admin and Support Features

i. Description and Priority

Administrators can manage users, monitor transactions, and handle support requests. Priority: Medium.

ii. Functional Requirements

REQ-4.5.1: The system shall provide an admin dashboard showing key metrics (active users, transaction volume, failed transactions).

REQ-4.5.2: Administrators shall be able to view, search, and temporarily suspend user accounts. REQ-4.5.3: The system shall allow administrators to review and resolve transaction disputes. REQ-4.5.4: All administrative actions shall be logged in an audit trail.

5. Other Nonfunctional Requirements

a. Performance Requirements

PERF-1: The system shall respond to user interface actions (page load, form submission) within 2 seconds under normal load. PERF-2: Fund transfer and payment confirmation shall complete within 5 seconds under normal network conditions.

PERF-3: The system shall support at least 1,000 concurrent users without significant degradation of response time. PERF-4: Database queries for transaction history (last 30 days) shall return results within 1 second.

b. Safety Requirements

SAFE-1: The system shall never allow a wallet balance to become negative.

SAFE-2: Critical financial operations shall be protected by confirmation steps and secondary authentication to prevent

accidental transfers. SAFE-3: The system shall maintain complete audit logs of all financial transactions for regulatory and dispute resolution purposes.

c. Security Requirements

SEC-1: All communication between client and server shall be encrypted using TLS 1.2 or higher. SEC-2: Passwords shall be stored using a strong one-way hashing algorithm (e.g., bcrypt or Argon2) with salt. SEC-3: Sensitive data such as bank account numbers and card details shall be encrypted at rest.

SEC-4: The system shall implement protection against common web vulnerabilities (SQL injection, XSS, CSRF) as per OWASP Top 10. SEC-5: Session tokens (JWT) shall have a limited lifetime and support secure logout/token revocation.

SEC-6: Administrative interfaces shall be accessible only to authorized personnel with role-based access control.

d. Software Quality Attributes

- Reliability: The system shall achieve 99.5% uptime excluding planned maintenance. • Usability: A first-time user should be able to complete registration and first top-up within 10 minutes with minimal assistance.
- Maintainability: Code shall be modular, well-documented, and follow consistent coding standards.
- Portability: The backend shall be deployable on major cloud platforms (AWS, Azure, GCP) and on-premise Linux servers.
- Scalability: Architecture shall support horizontal scaling of application servers and database read replicas.

e. Business Rules

BR-1: Only KYC-verified users may perform full wallet operations (top-up beyond a basic limit, withdrawals, and high-value transfers).

BR-2: Daily and monthly transaction limits shall be enforced as per applicable PPI regulations. BR-3: A user may hold only one primary wallet account per registered mobile number. BR-4: Administrators with appropriate roles may suspend accounts suspected of fraudulent activity pending investigation.

6. Other Requirements

Database Requirements: The system shall use a relational database with proper normalization, indexing on frequently queried fields (userID, transaction date, status), and regular backup schedules.

Internationalization: The initial version shall support English. Future versions may add multi-language support (Hindi and other regional languages).

Legal / Compliance: The system design shall consider data protection principles (purpose limitation, data minimization) and applicable financial regulations for digital wallets.

Logging & Monitoring: Application and security logs shall be retained for a minimum period as required by regulation (typically 5– 8 years for financial records). Appendix A: Glossary

Term	Definition
DWMS	Digital Wallet Management System – the software product specified in this document.
KYC	Know Your Customer – process of verifying the identity of a user.
OTP	One-Time Password – a temporary code sent for authentication.
P2P	Peer-to-Peer – direct transfer of funds between two individual users.
PPI	Prepaid Payment Instrument – regulatory category covering digital wallets.
QR Code	Quick Response code used for initiating payments by scanning.
TLS	Transport Layer Security – cryptographic protocol for secure communication.
JWT	JSON Web Token – compact token format used for authentication.
UPI	Unified Payments Interface – real-time payment system in India.

Appendix B: Analysis Models

The following analysis models are recommended to be developed during the design phase (not included in this version of the SRS):

- Use Case Diagrams for major actors (Registered User, Merchant, Administrator). • Entity-Relationship Diagram for the database schema (User, Wallet, Transaction, BankAccount, etc.).
- Sequence Diagrams for critical flows: Registration, Login, Add Money, P2P Transfer. • State Transition Diagram for Transaction status (Initiated → Processing → Success / Failed).

Appendix C: To Be Determined List

The following items require further clarification or decision:

6. Exact third-party payment gateway and banking partners to be integrated. 7. Final daily/monthly transaction limits (pending regulatory confirmation for the target market).